GLOBAL SUPERSTORE

PROJECT DESCRIPTION

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CONTENS

1 BUSINESS DESCRIPTION

1.1 BUSINESS BACKGROUND

Retail trade is one of the most dynamic and competitive business domains in the global economy.

Modern retail companies operate across multiple markets and regions, offering a wide range of products through both online and offline sales channels. Customers today expect fast delivery, competitive pricing, and a personalized shopping experience, which significantly increases the complexity of retail operations.

The chosen business domain for this project is a global retail company that sells consumer goods across multiple product categories, including technology, office supplies, and furniture. The company operates through two main sales channels: a modern online web store and a legacy offline sales system (physical stores managed through an ERP system). Each channel generates large volumes of transactional data related to orders, customers, products, geography, and employees.

1.2 PROBLEMS BECAUSE OF POOR DATA MANAGEMENT

Fragmented data across regional and legacy systems hinders retail performance. Without a centralized repository, the business suffers from inconsistent product data, manual reporting errors, and an incomplete view of customer behavior and sales channels. These inefficiencies prevent data-driven decision-making, undermining profitability and market competitiveness.

1.3 BENEFITS FROM IMPLEMENTING A DATA WAREHOUSE

A properly designed DWH enables historical analysis, improves data quality, and supports complex analytical queries.

By implementing a Data Warehouse, the business will be able to answer questions such as:

- What are the total sales and profits by product category and subcategory?
- How do online and offline sales channels compare in terms of revenue and cost?
- Which regions and markets generate the highest sales and profitability?
- How do shipping costs and discounts impact overall profit?
- Which customers and segments contribute the most to revenue?

Further data processing and analysis will also allow the company to:

- Identify sales trends over time
- Optimize inventory and warehouse operations
- Improve pricing and discount strategies
- Support strategic and tactical business decisions

1.4 DATASETS DESCRIPTION

Original dataset was Global Superstore (Kaggle) and using its base the following 2 datasets were created: online-web-store and offline-legacy-erp

- Split principle:
  Online = Modern Web Store
  Offline = Legacy ERP (In-store)
- Added columns:
  Online: Device_Type, Payment_Method, Session_ID, Channel, Product supplier

  Offline: Cashier Names and contact information, Store_Type and its location details, Product supplier, Channel

- Second dataset generated by:
  - Rule-based splitting (removing Store and Employee details, adding attributes specific for only online itself)
  - Attribute transformation
  - ID transformation (WEB- / ERP-)
- 1million records achieved by:
  - Dataset duplication
  - Date shifting (+30/+60/+90 days, and +10years)
  - ID regeneration

1st Dataset – Online Sales System (Modern Web Store)

The first dataset represents sales transactions generated by a modern online retail platform.

Key entities:

- **Date**: Order Date, Ship Date
- **Customer**: Customer ID, Customer First Name, Customer Last Name, Customer Phone, Customer Segment
- **Geography***: Geo PostalCode, Geo City, Geo State, Geo Country, Geo Region, Geo Market
- **Product**: Product ID, Category, Subcategory, Product Name, Product Supplier

(**Hierarchy:** Category → Sub-Category → Product)

- **Channel / Session**: Channel, Device Type, Payment Method, Session ID

- **Sales and Cost Measures:** Sales, Quantity, Discount, Profit, Shipping Cost

2nd Dataset – Offline Sales System (Legacy ERP)

The second dataset represents sales transactions recorded in a legacy ERP system used by physical retail stores. Although it contains similar sales facts, the dataset differs in structure and attributes, reflecting the specifics of offline sales operations.

- **Date:** Order_Date, Ship_Date
- **Customer:** Customer_ID, Customer_FirstName, Customer_LastName, Customer_Phone, Customer_Segment
- **Geography*:** Geo PostalCode, Geo City, Geo State, Geo Country, Geo Region, Geo Market
  Hierarchy: Market → Region → Country → State → City
- **Store:** Store_ID, Store_Type, Store_Address, Store_phone, Store_City Store_State Store_Country
- **Product:** Product_ID, Category, Sub_Category, Product_Name Product_Supplier
- **Employee / Cashier:** Cashier_ID, Cashier_FirstNname, Cashier_LastNname, Cashier_Email, Cashier_phone
    - **Channel - Distinguishes offline sales from other sales channels.**

*The Geography dimension represents the delivery address location for online orders and the customer purchase location for offline orders. It does not represent store physical location, which is stored separately in the Store dimension.

So the difference between 2 datasets:

Feature	Online Web Store	Offline Legacy ERP
Channel	Online	Offline
Order Source	Web Platform	POS System
Device info	Yes	No
Cashier info	No	Yes
Store info	Virtual store	Physical store
Payment type	Digital	Cash/Card

1.5 GRAIN / DIM / FACT

Grain description

Step 1 – Business process

The selected business process is **Global Retail Sales Transactions** across two source systems:

- Modern Web Store (Online)
- Legacy ERP (Offline)

The process records customer purchases of products, including sales and cost-related measures.

Step 2 – Declare the grain

One row represents one Sale of the product in a single sales transaction (order and shipped), on a specific date, to a specific customer, in a specific location, by specific cashier via a specific sales channel (online or in-store).

Step 3 – Identify the dimensions

This grain was selected as it allows flexible aggregation of Sales by answering to these questions:

Who bought it? → Customer

What was sold? → Product

Who sold it? → Employee

Where was it sold? → Geography

How it was sold? → Sales channel

Supports future analytical requirements such as:

- Daily / monthly sales trends
- Profitability by product and region
- Online vs Offfline performance comparison

Step 4 – Identify the facts

Although the data originates from two different source systems, the identified facts remain consistent. Because both systems record monetary and quantity values at transaction level and differences exist only in technical attributes, not in measures

Column name	Description	Data Type
Field Name 1	<description>	Int
Filed Name N	<description>	Text

FACT_SALES

Stores measurable sales and cost data at transactional grain.

Column Name	Description	Data Type
Sales_ID	Surrogate fact key	Int

Date_ID	Foreign key to Dim_Date	Int
Customer_ID	Foreign key to Dim_Customer	Text
Product_ID	Foreign key to Dim_Product	Text
Geography_ID	Foreign key to Dim_Geography	Int
Store_ID	Foreign key to Dim_Store	Text
Employee_ID	Foreign key to Dim_Employee	Text
Quantity	Quantity sold	Int
Sales_Amount	Total sales amount	Decimal
Discount	Discount applied	Decimal
Profit	Profit amount	Decimal
Shipping_Cost	Cost of shipping	Decimal
Product_Cost	Derived product cost	Decimal

Example with filled data

Sales_ID	Date_ID	Customer_ID	Product_ID	Geography_ID	Store_ID	Employee_ID	Quantity	Sales_Amount	Profit
1	20240115	KW-657086	OFF-FA-6196	10	STORE-031	EMP-2105	2	143.10	5.70

DIM_PRODUCT

Stores product details and product hierarchy.

Column Name	Description	Data Type
Product_ID	Natural product identifier	Text
Product_Name	Product name	Text
Sub_Category	Product sub-category	Text
Category	Product category	Text

Example with filled data

Product_ID	Product_Name	Sub_Category	Category
OFF-FA-6196	Memorex USB Flash Drive	Accessories	Technology

DIM_ CUSTOMER

describes who is buying.

Column Name	Description	Data Type
Customer_ID	Natural customer identifier	Text
Customer_Name	Full customer name	Text
Segment	Customer segment	Text

Example with filled data

Customer_ID	Customer_Name	Segment
KW-657086	Maria Gonzalez	Consumer

DIM_ DATE

supports time-based analysis

Column Name	Description	Data Type
Date_ID	Surrogate key for date	Int
Full_Date	Full calendar date	Date
Day	Day of month	Int
Month	Month number	Int
Month_Name	Month name	Text
Quarter	Calendar quarter	Int
Year	Calendar year	Int

Example with filled data

Date_ID	Full_Date	Day	Month	Month_Name	Quarter	Year

20240115	2024-01-15	15	1	January	1	2024

DIM_ GEOGRAPHY

Stores geographical information and location hierarchy.

Column Name	Description	Data Type
Geography_ID	Surrogate key	Int
Country	Country name	Text
State	State or province	Text
City	City name	Text
Postal_Code	Postal code	Text
Region	Sales region	Text
Market	Market name	Text

Example with filled data

Geography_ID	Market	Region	Country	State	City	Postal_Code
101	Africa	Central Africa	Cameroon	Centre	Yaounde	00237

DIM_ STORE

Stores store and channel-related information.

Column Name	Description	Data Type
Store_ID	Store identifier	Text
Store_Address	Store address	Text
Store_Type	Physical store type (Offline only)	Text
Store_phone	Phone number to connect	text

Example with filled data

Store_ID	Store_Address	Store_Type	Store_phone
STORE-031	12 Main St	Mall	+1-474-951-1828

DIM_ Cashier

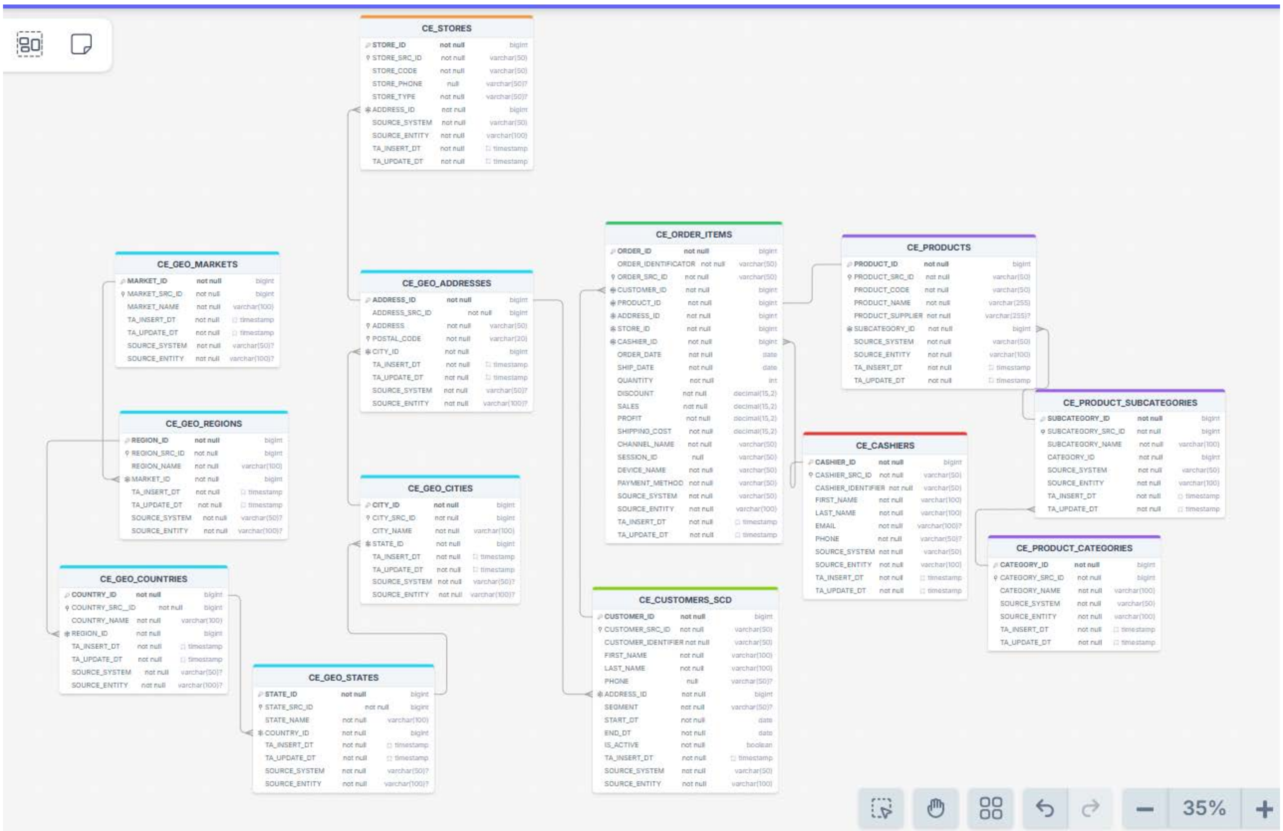
describes who is buying.

Column Name	Description	Data Type
Cashier_ID	Surrogate employee key	Text
Cashier_Name	Salesperson name	Text
Cashier_Email	Salesperson mail	Text
Cashier_phone	Salesperson phone	text

Example with filled data

Cashier_ID	Cashier_Name	Cashier_Email	Cashier_phone
EMP-2105	Preecha Metharom	Preecha Metharom@store.com	+1-377-354-6678

2 BUSINESS LAYER 3NF



The design followed a accurate normalization process to transition from source datasets (online-web-store and offline-legacy-erp) into a centralized 3NF structure.

Step 1: Data Unification and Entity Identification

We identified core entities present in both datasets. To maintain a "single source" entities were unified into single tables rather than maintaining separate tables per source system.

Step 2: Normalization to 3rd Normal Form (3NF)

To eliminate data redundancy and transitive dependencies, we decomposed flat structures into hierarchical tables:

Geography: Split into a 6-level hierarchy (Market → Region → Country → State → City → Address).

Products: Split into a 3-level hierarchy (Category → Subcategory → Product).

Sales: Divided into Header (CE_ORDERS) and Line Items (CE_ORDER_ITEMS) to ensure that order-level attributes (like dates and customers) are stored only once per transaction.

Step 3: Application of Naming Conventions

Step 4: Technical Meta

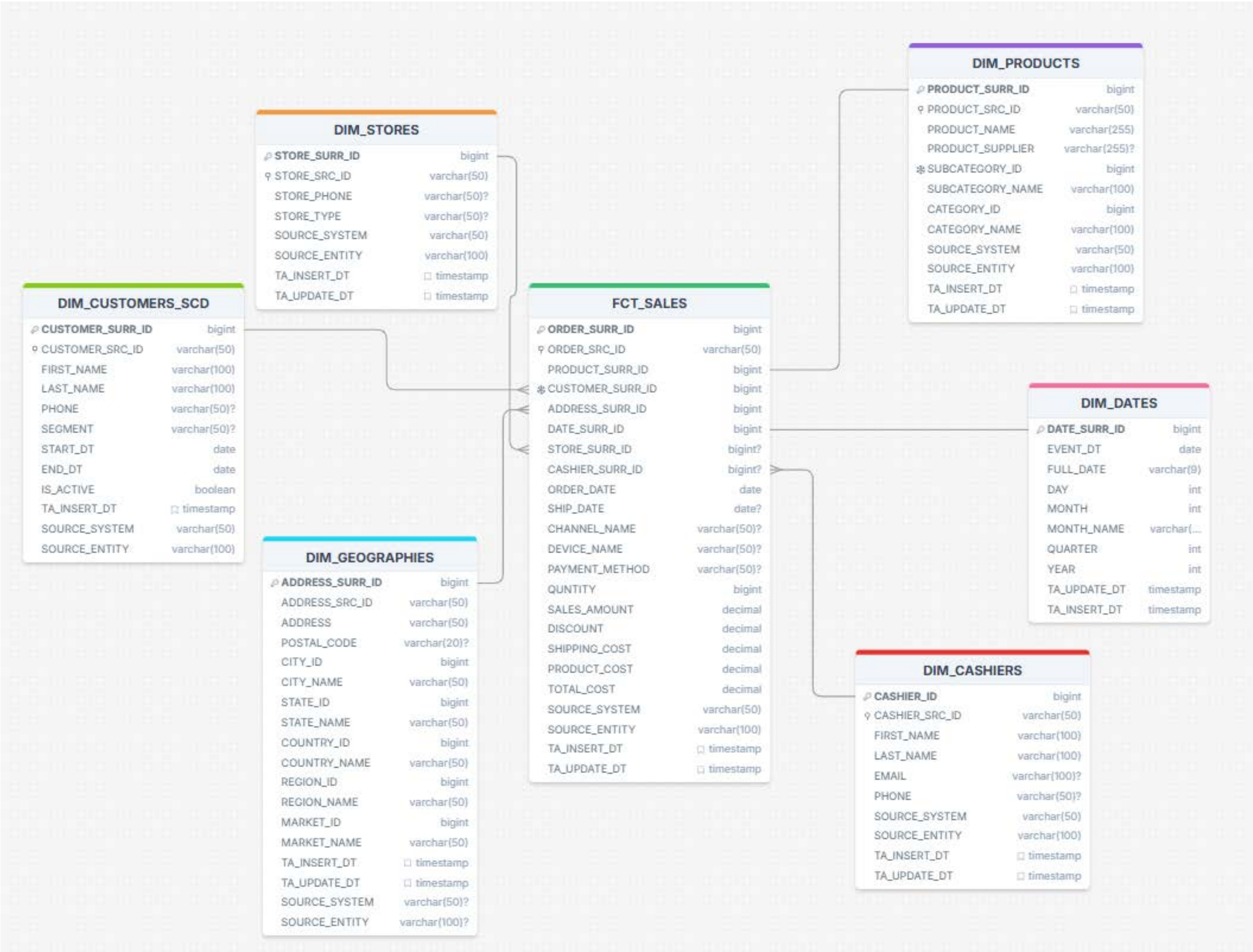
To ensure auditability every table includes:

- **Source:** SOURCE_SYSTEM, SOURCE_ENTITY.
- **Technical Attributes:** TA_INSERT_DT and TA_UPDATE_DT to track when records entered the warehouse.

The Customer entity was implemented by SCD Type 2 implementation because there are:

- High likelihood of attribute changes (phone, address, segment, name, etc.)
- Business relevance of historical tracking
- Analytical importance of customer history

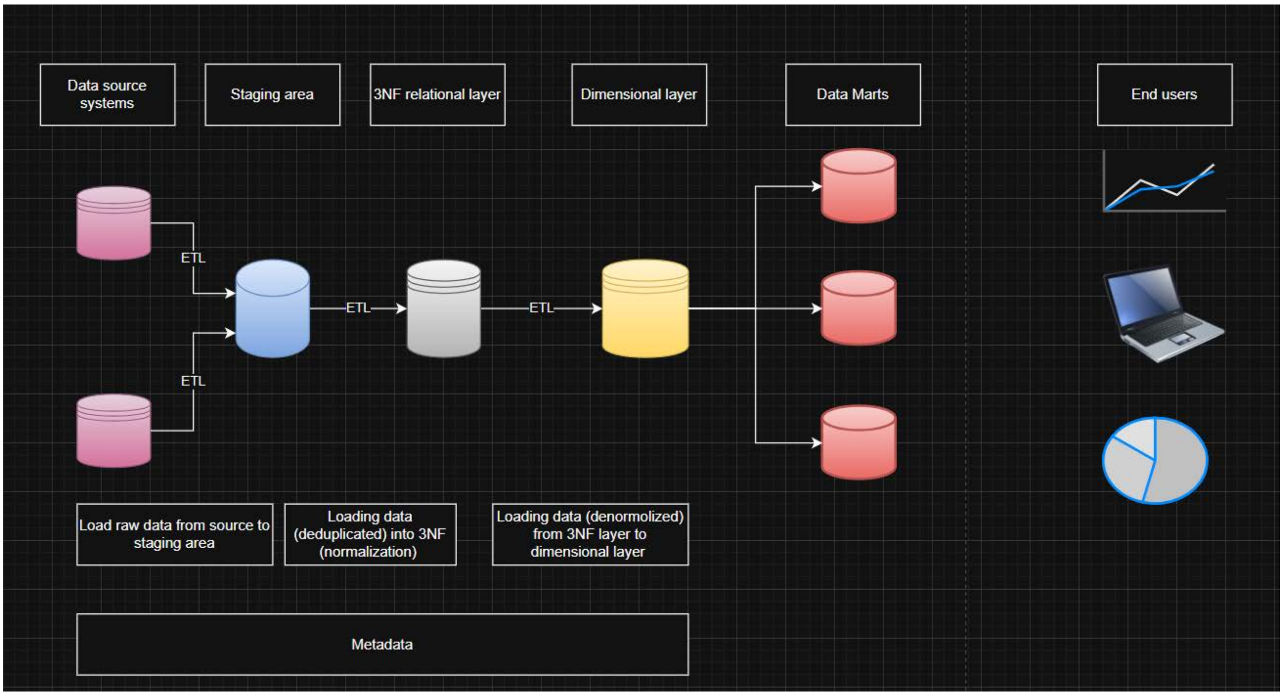
3 BUSINESS LAYER DIMENSIONAL MODEL



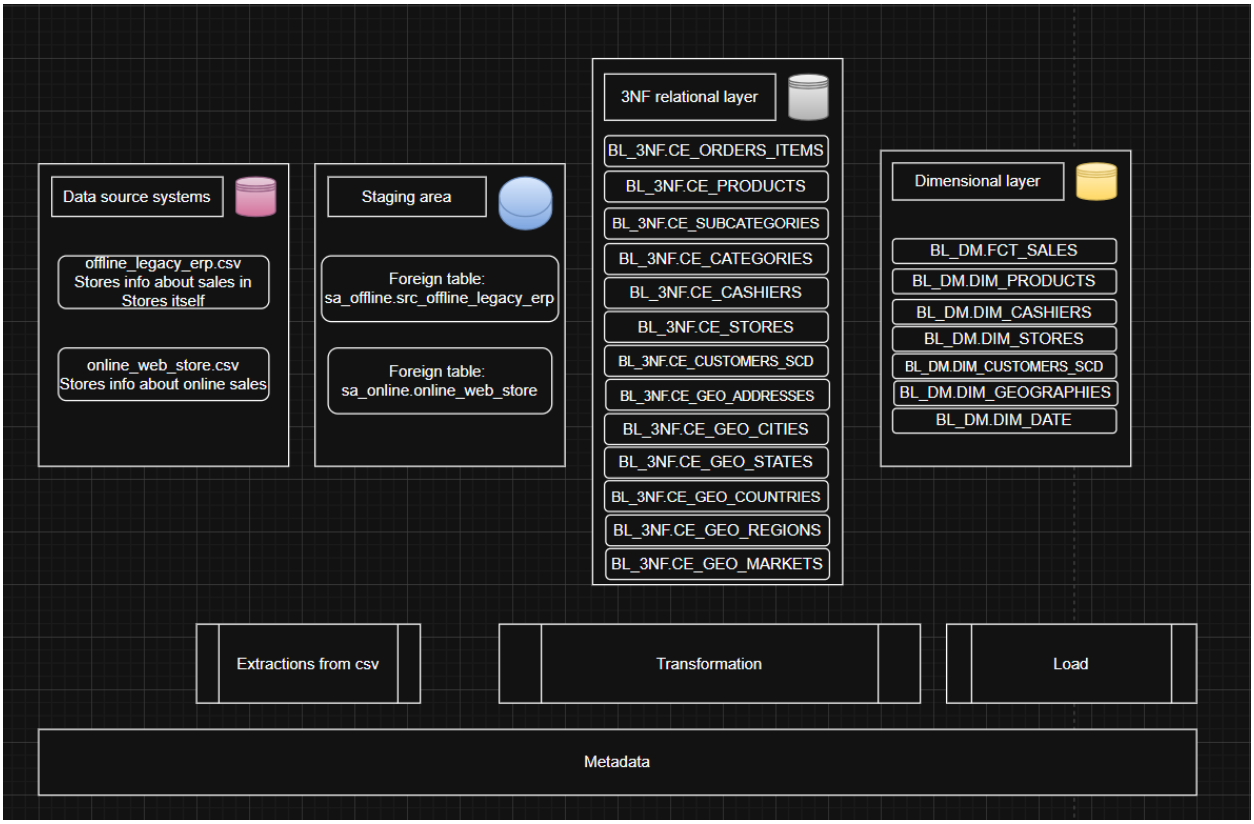
The dimensional layer is built as a star schema where normalized hierarchical structures from the 3NF layer are denormalized into single dimension tables (Geography, Product), and business measures are stored in a central fact table at the order-item grain (One row represents one sales transaction line for one product sold in a single order on a specific date). Fact table stores these measures:

- SALES
- QUANTITY
- DISCOUNT
- PROFIT
- SHIPPING_COST
- TOTAL_COST (calculated metric which is equal to SHIPPING_COST + (SALES - PROFIT))

4 LOGICAL SCHEME



5 DATA FLOW



6 FACT TABLE PARTITIONING STRATEGY

The table `FCT_SALES` is partitioned by year (`RANGE` by `date_id`), because most business analytical queries are filtered by period, and the annual partition remains compact with the current volume (~250-330 thousand rows/year) without an excessive number of small partitions. The detailed justification is in `sql/04_partitioning.sql'.

Working ETL Pipeline

`etl/etl_pipeline.py ` is not just a design on paper, but a working script on Python (pandas), which:

1. Reads both CSV sources and brings them to a common structure
2. Builds all dimensions (Date, Product, Customer, Geography, Store, Cashier)
3. Builds a fact table with the calculated Total_Cost field
4. Uploads the result to SQLite (`output/retail_dwh.db')